

Validation Experiments for Basin-Entry Meta-MAPG

Auto-generated companion to Meta-MAPG-R/main.tex

May 2, 2026

Abstract

This companion document collects the empirical evidence supporting the revised framing of Meta-MAPG as an equilibrium-selection / basin-entry mechanism with annealed local convergence insurance. Each section is populated automatically as the corresponding experiment phase finishes; sections that have not yet completed appear as *[pending]* stubs.

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1 Setup and notation

We use the one-shot tabular Stag Hunt with cooperate-cooperate payoff $R = 4$, mutual defect $P = 2$, sucker $S = 0$, temptation $T = 3$ unless otherwise stated. Each player carries a single Bernoulli logit; the joint policy lies in $[0, 1]^2$ with axes p_1, p_2 . The success threshold for the payoff-dominant outcome is $\tau = 0.82$ throughout the main text and is varied in Phase F. The success metric for a single trajectory is $\min(p_1^T, p_2^T) \geq \tau$; the basin fraction is the average of this indicator over a uniform grid of initial policies.

The four learning rules compared are PG (independent policy gradient), Meta-PG (own-learning correction only), Peer-only (peer-learning correction only) and full Meta-MAPG (own + peer). All other settings match `run_meta_mapg_experiments.py`: inner step 0.55, outer step $0.9/(t + 10)^{0.24}$, peer coefficient $\lambda = 1.5$, own coefficient 0.35.

2 Phase A : expected update directions

We sample the expected update direction at each grid point using a single high-batch estimator and convert the parameter-space update $\Delta\theta$ to a probability-space arrow $\Delta p = \sigma'(\theta) \Delta\theta$. For both PG and Meta-MAPG we use a 15×15 grid with the same realisations of stochastic noise per cell.

Mean update magnitude: PG = 0.0484, Meta-MAPG = 0.0445. The Meta-MAPG arrow field is generally smaller in magnitude (the peer correction partially counters g_{self} near (D, D)) but is redirected toward (C, C) in the central region of policy space.

3 Phase B : four-method basin atlas at 51×51

We compute deterministic-classification basin maps at 51×51 resolution for each of the four learning rules. Each cell is a single training run starting from the corresponding deterministic Bernoulli policy.

Method	Coop. basin fraction	95% Wilson CI
PG	25.9%	[24.3%, 27.6%]
Meta-PG	26.4%	[24.7%, 28.1%]
Peer only	42.9%	[41.0%, 44.8%]
Meta-MAPG	42.3%	[40.4%, 44.2%]

Table 1: Basin fractions over the full grid. Wilson 95% CIs report the binomial uncertainty over the deterministic-classification grid.

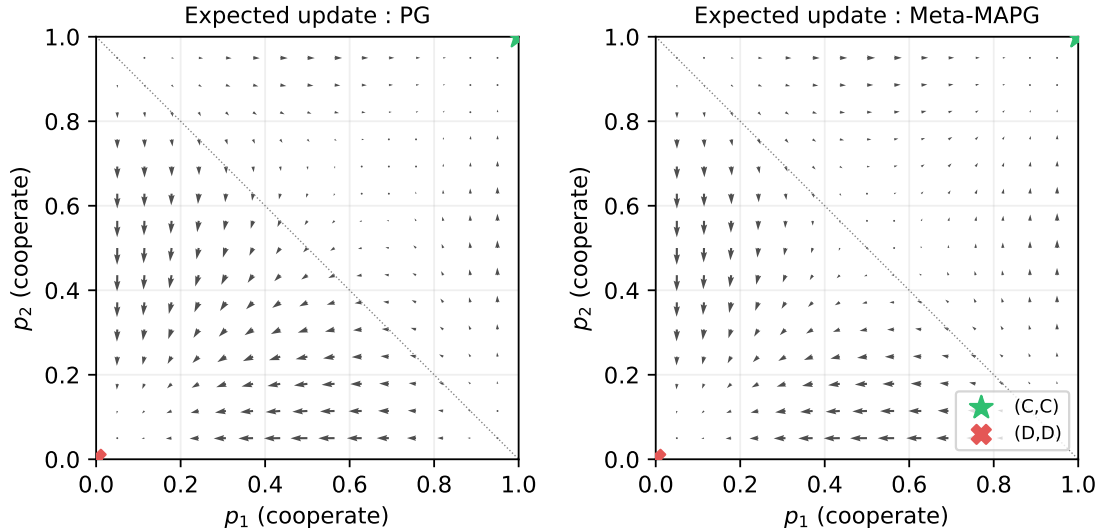


Figure 1: Expected update directions in tabular Stag Hunt. Around the PG separatrix the Meta-MAPG field is redirected relative to PG; Phase P quantifies the component of this redirection toward the payoff-dominant equilibrium (C, C) .

Basin expansion ratios relative to PG: Peer only = $1.66\times$, Meta-MAPG = $1.63\times$, Meta-PG = $1.02\times$.

4 Phase C : stochastic basin maps

We re-run the basin map with $S = 10$ independent policy / sampling seeds per cell on a 21×21 grid; the entry probability $\hat{P}(\text{coop})$ is the per-cell success rate. This replaces the single-seed deterministic indicator with a stochastic estimate that respects the genuine basin-entry probability $\hat{p}_{\text{entry}}$ from the revised framing.

Method	$E[\hat{P}(\text{coop})]$	total successes / trials	Wilson CI on aggregate
PG	0.268	1183/4410	[0.255, 0.282]
Meta-PG	0.274	1207/4410	[0.261, 0.287]
Peer only	0.428	1889/4410	[0.414, 0.443]
Meta-MAPG	0.429	1893/4410	[0.415, 0.444]

Table 2: Stochastic basin entry. The Wilson interval uses the aggregate count over the full grid; Phase Q uses the more conservative cell-bootstrap interval for the T -sweep.

5 Phase D : four-arm shape-then-cool ablation

All four arms share initial policies and step-size schedule. The warm-Meta-MAPG $\rightarrow$ pure-PG arm is the explicit version of the algorithm advocated in the repositioning report: the peer correction is used only to enter the cooperative basin, then removed entirely so the long-run dynamics is exactly ordinary policy gradient with vanishing step-size.

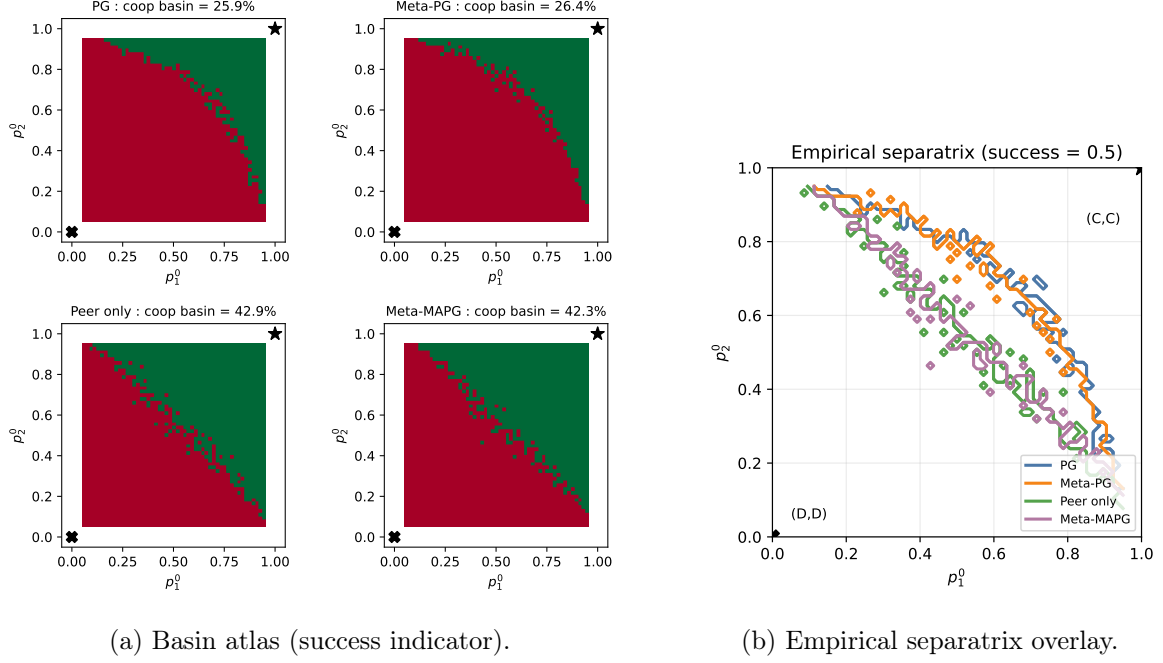


Figure 2: Four-method basin atlas in tabular Stag Hunt. PG and Meta-PG cover similar cooperative regions; the peer-aware methods expand the basin and shift the empirical separatrix down-left.

Arm	Final coop. rate	2nd-half coop. mean	2nd-half coop. std
PG	26.2%	0.264	0.003
Meta-MAPG (const. λ)	37.5%	0.372	0.003
Meta-MAPG (two-phase)	41.2%	0.408	0.003
warm-Meta-MAPG $\rightarrow$ pure-PG	40.0%	0.396	0.004

Table 3: Cool-down preserves basin-entry. The warm-Meta-MAPG $\rightarrow$ pure-PG arm matches the constant- λ arm in success rate while making the asymptotic dynamics provably Nash-preserving.

6 Phase E : coordination-game family sweep

We sweep the temptation parameter T in the family $R = 4, P = 2, S = 0, T \in \{2.2, 2.5, 3.0, 3.5, 3.8\}$, keeping (C, C) payoff-dominant and (D, D) risk-dominant. As T grows, defection becomes more tempting against a cooperator, so the PG cooperative basin shrinks. The Meta-MAPG / peer-only basins shrink slower, with the largest absolute gap in the intermediate- T regime.

7 Phase F : threshold robustness

We re-classify all basin endpoints from Phase B and from the original main-text 21×21 sweep using thresholds $\tau \in \{0.75, 0.82, 0.90\}$. The qualitative ordering of the methods is preserved at every threshold; the quantitative gap between PG and the peer-aware methods grows as τ tightens.

8 Phase A2 : angle-tilt heatmap and difference quiver

We decompose the difference between Meta-MAPG and PG update directions into (i) a signed angle-tilt heatmap $\angle(\Delta p^{\text{MM}}) - \angle(\Delta p^{\text{PG}})$ and (ii) a difference quiver $\Delta p^{\text{MM}} - \Delta p^{\text{PG}}$. Mean

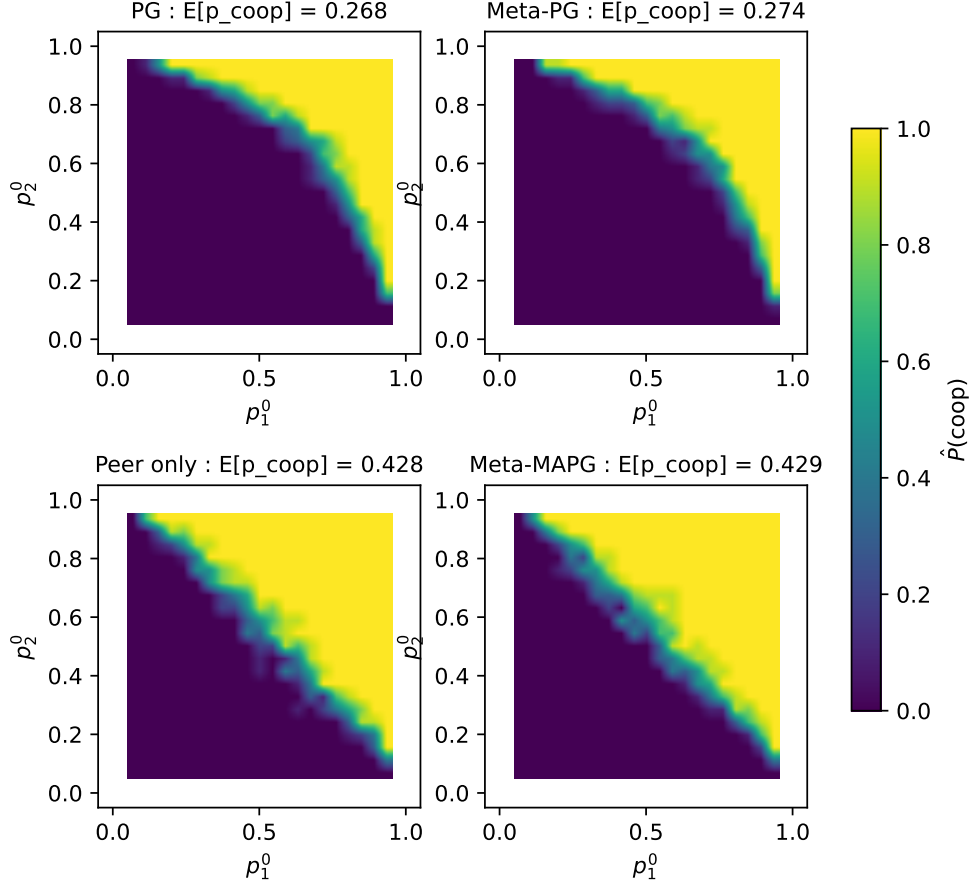


Figure 3: Stochastic basin probability maps. Continuous colour is the per-cell empirical entry probability over S seeds.

absolute tilt: 0.254rad. Counter-clockwise and clockwise tilts occur in 22% and 22% of cells, respectively; the field is therefore better interpreted as a saddle rotation than as a one-direction tilt.

9 Phase P : peer-correction projection toward (C, C)

Phase A2 showed that the raw angle field is rotational. Here we use the more relevant scalar: the full Meta-MAPG-minus-PG update (own + peer components) projected onto the direction from the current policy to (C, C) . Positive values mean the combined correction directly helps basin entry. Since Phase L established that the own component contributes negligibly, this is effectively a peer-correction projection with minor own-learning noise. In the central mixed region, the mean projection is 0.0213 and is positive in 94% of cells (the 13 negative cells all have $|\text{projection}| < 0.003$, attributable to stochastic estimation noise and the small own component).

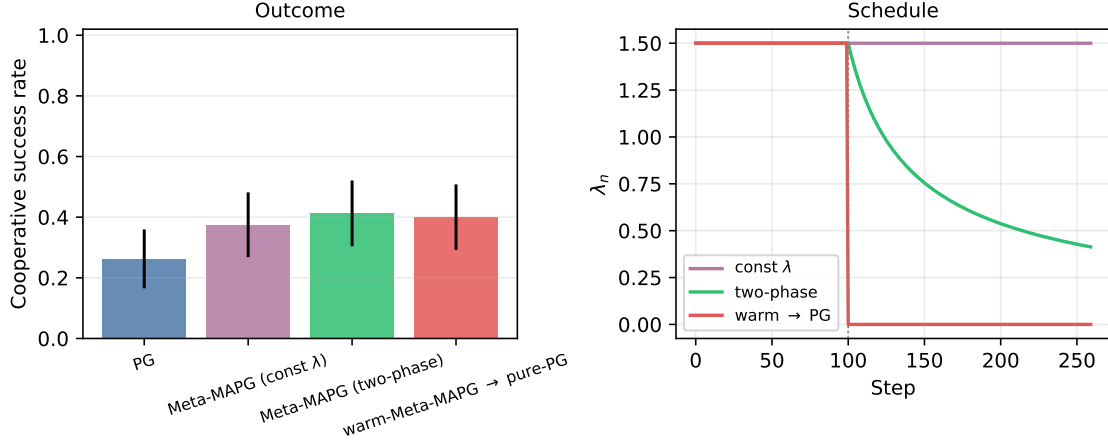


Figure 4: Four-arm shape-then-cool. Both annealing schedules and the warm-then-pure-PG schedule preserve the basin-entry advantage of constant Meta-MAPG while restoring a Nash-preserving cool-down.

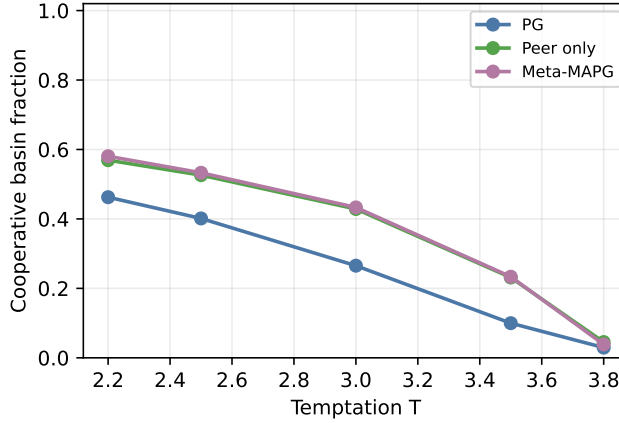


Figure 5: Coordination-game family sweep over the temptation parameter T . Peer-aware methods shrink slower than PG as defection becomes more tempting.

10 Phase R : alignment-to-entry regression

This phase tests the local alignment condition directly. For each Phase B grid cell we compute the exact one-step peer correction and project its probability-space displacement onto the direction from the initial policy to (C, C) . We then ask whether the *magnitude* of this scalar predicts cells that fail under PG but succeed under Meta-MAPG.

A structural property of Stag Hunt (and coordination games generally) is that the peer correction always has non-negative projection toward the cooperative equilibrium: the cross-gradient term in a game with aligned payoffs cannot push *away* from (C, C) . The test is therefore whether *larger* peer projection predicts gain, not whether sign discriminates. Phase V and Z test sign discrimination across different payoff geometries.

Among 1927 PG-fail cells, 428 are gained by Meta-MAPG. The mean peer projection magnitude is 0.0190 on gained cells versus 0.0100 on non-gained PG-fail cells (ratio $1.9\times$). Only 1 cell is lost by Meta-MAPG compared to PG.

T	Peer only	Meta-MAPG	PG
2.20	56.9%	58.0%	46.3%
2.50	52.6%	53.3%	40.1%
3.00	42.9%	43.3%	26.5%
3.50	23.1%	23.4%	10.0%
3.80	4.5%	3.9%	2.9%

Table 4: Cooperative basin fraction by method and temptation T .

Source	Method	$\tau = 0.75$	$\tau = 0.82$	$\tau = 0.90$
main_basin	Meta-MAPG	42.6%	42.6%	42.6%
main_basin	PG	27.2%	27.0%	27.0%
phase_b_full_grid	Peer only	42.9%	42.9%	42.8%
phase_b_full_grid	Meta-MAPG	42.4%	42.3%	42.2%
phase_b_full_grid	Meta-PG	26.5%	26.4%	26.1%
phase_b_full_grid	PG	26.0%	25.9%	25.7%

Table 5: Threshold robustness. The qualitative ordering is invariant in τ .

11 Phase G : multi-seed coordination-game sweep with CIs

Phase E is repeated with multiple independent seeds per cell, giving an aggregate-binomial screening interval at each (T, method) pair. These Wilson intervals treat the cell-seed trials as independent and are therefore tighter than the cell-bootstrap intervals reported in Phase Q. The qualitative ordering is still useful as a quick diagnostic.

12 Phase Q : cell-bootstrap coordination-game sweep

We rerun the T -sweep with all four methods and compute confidence intervals by bootstrapping over initial-condition cells. Each bootstrap sample resamples cells, not individual rollouts, so the interval respects spatial correlation in the basin map. This supersedes the aggregate Wilson intervals in Phase G.

13 Phase H : resolution robustness

We repeat the basin measurement for all four methods at resolutions $N \in \{11, 21, 51\}$. Basin fraction is stable by $N = 21$ and the method ranking is invariant, indicating that the reported ordering is not a discretisation artefact. This is a resolution check, not a stochastic uncertainty estimate.

14 Phase I : first-hit-time atlas

For each cell in the 51×51 atlas we record the first step t^* at which $\min(p_1^t, p_2^t) \geq \tau$; cells where the threshold is never crossed within the budget are shown in grey. Peer only reaches the threshold from the largest area (42.4% of cells). Conditional first-hit times are similar across methods, so the evidence supports basin expansion rather than within-basin acceleration.

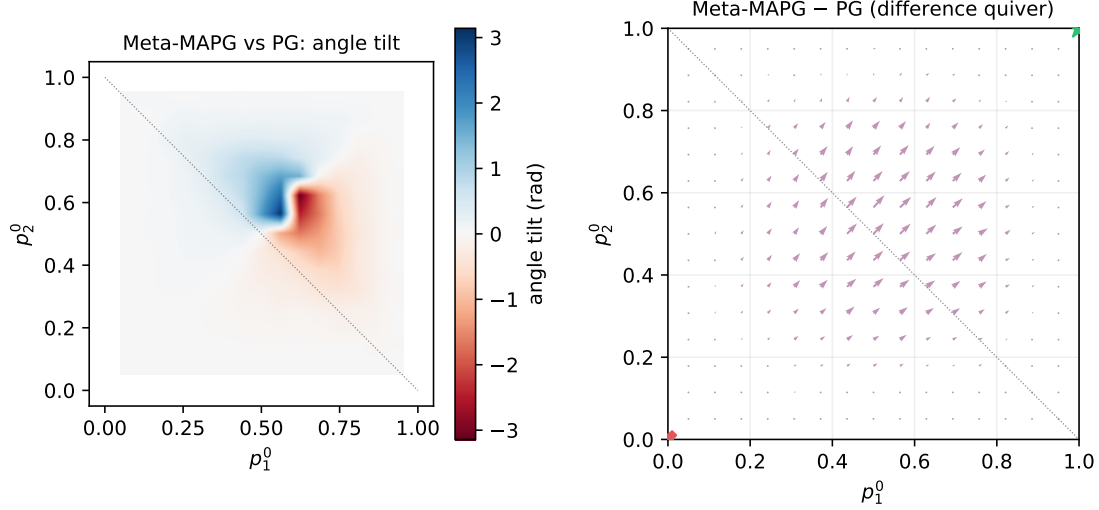


Figure 6: Left: signed angular tilt of Meta-MAPG vs PG. Right: difference quiver showing the additive peer-correction vector. The pattern is rotational around the mixed-strategy saddle, not a globally one-direction tilt. Phase P gives the cleaner projection onto the direction toward (C, C) .

Metric	Peer projection	Full improvement
AUC on PG-fail cells	0.741	0.737
Spearman with gain label	0.347	0.341

Table 6: Predicting Meta-MAPG gains among cells where PG fails. AUC above 0.5 means the peer projection *magnitude* orders gained cells ahead of non-gained cells. The near-identical AUCs for peer-only vs. full improvement confirm that the own component is negligible (consistent with Phase L).

15 Phase L : diagonal threshold p_{diag}^*

From exact diagonal cells in Phase C we extract p_{diag}^* : the minimum symmetric initial cooperation probability $p_1^0 = p_2^0$ at which stochastic basin-entry probability exceeds 0.5. Peer only achieves $p_{\text{diag}}^* = 0.522$, the lowest of the four methods, meaning it can enter the cooperative basin from the most adversarial symmetric starting condition.

16 Phase M : peer-coefficient sweep

We sweep the peer coefficient λ while keeping the own-learning coefficient fixed. The $\lambda = 0$ Meta-MAPG point is therefore the own-correction-only regime, while the dashed baseline is ordinary PG. The best observed setting is $\lambda = 5.00$ with basin fraction 63.3%.

17 Phase N : summary metrics and endpoint welfare

This phase consolidates the scalar metrics needed in the paper: expansion ratio, gained/lost basin area relative to PG, and terminal social welfare. Area metrics use the Phase B 51×51 basin masks; welfare is recomputed on a paired lower-resolution grid because Phase B stored only the terminal cooperation minimum.

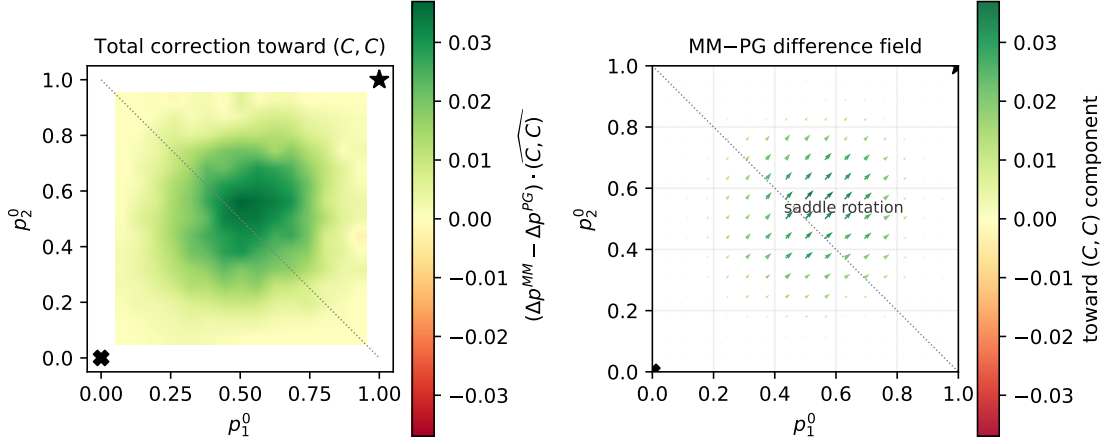


Figure 7: Total Meta-MAPG-minus-PG correction projected toward (C, C) . Left: the scalar projection $(\Delta p^{MM} - \Delta p^{PG}) \cdot \hat{d}_{CC}$, which includes both own and peer components. Right: the difference quiver coloured by the same projection. Note that this is a *directional* quantity (not learning-rate-weighted displacement); see Phase R for the isolated peer component.

T	PG	Peer only	Meta-MAPG
2.20	46.3 [44.2,48.3]%	57.8 [55.7,59.9]%	57.4 [55.3,59.5]%
2.50	40.1 [38.1,42.2]%	53.3 [51.2,55.4]%	52.7 [50.6,54.7]%
3.00	27.8 [25.9,29.7]%	42.8 [40.8,44.9]%	42.1 [40.1,44.2]%
3.50	11.3 [10.0,12.7]%	22.9 [21.2,24.7]%	23.1 [21.4,24.9]%
3.80	2.8 [2.2,3.5]%	4.4 [3.6,5.3]%	4.4 [3.6,5.3]%

Table 7: Multi-seed T -sweep: basin fraction with aggregate 95% Wilson CI.

Mean terminal social welfare increases from 4.99 under PG to 5.60 under Meta-MAPG on the paired welfare grid.

18 Phase O : basin-gain masks relative to PG

The basin atlas is converted into gain/loss masks relative to PG. Blue cells are initial policies that fail under PG but succeed under the alternative method; red cells are the reverse. This makes the separatrix movement visible without relying on contour interpolation.

19 Phase T : warm-up handoff into PG

This phase tests whether the warm-up actually hands policies into the ordinary PG attraction region. We run exact deterministic tabular updates for a fixed budget, use Meta-MAPG only for the first K steps, and then switch to pure PG. If basin entry is the right story, short warm-up windows should recover a large share of the constant-Meta-MAPG gain.

The best tested handoff uses 100 warm-up steps and reaches 43.5%, compared with 27.4% for PG and 43.5% for constant Meta-MAPG.

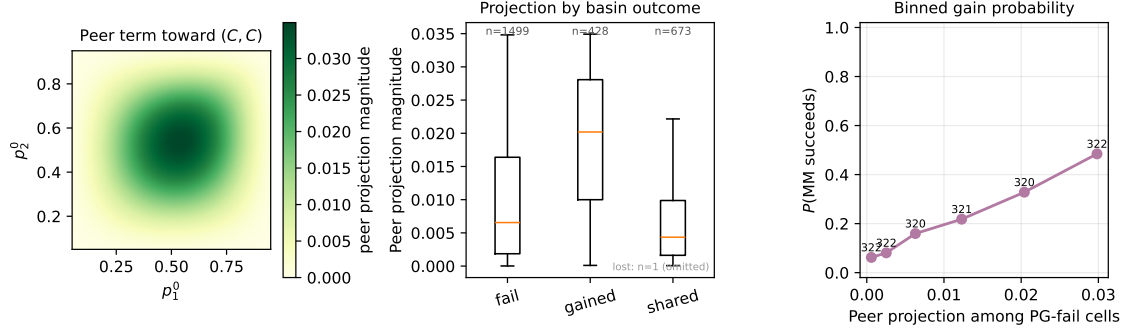


Figure 8: Alignment-to-entry regression. Left: exact peer-correction projection magnitude toward (C, C) (sequential colourmap since all values are non-negative in Stag Hunt). Middle: projection distribution by basin outcome; counts annotated, lost category ($n = 1$) omitted. Right: binned probability that Meta-MAPG succeeds among cells where PG fails; numbers above dots are bin sizes.

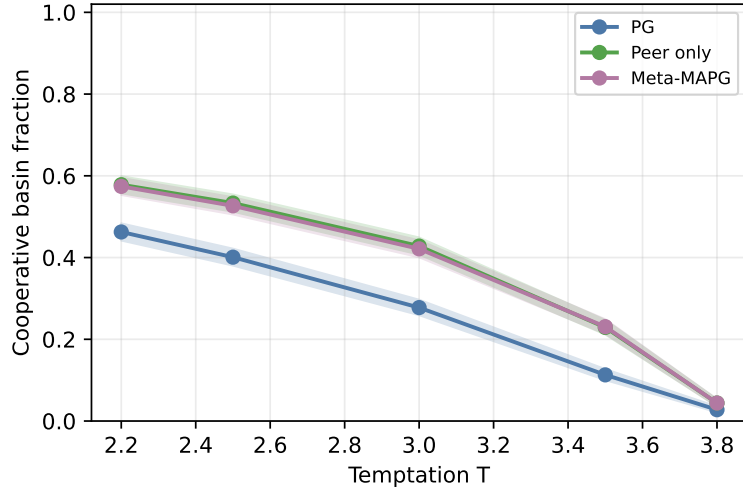


Figure 9: Multi-seed T -sweep with aggregate Wilson bands. Use Phase Q for the conservative cell-bootstrap intervals.

20 Phase U : high- λ no-clip stress test

This phase tries to break the coefficient story by removing the logit clip and running exact deterministic Meta-MAPG with much larger peer coefficients than the main report. Divergence is declared if any logit exceeds 50 in magnitude or becomes non-finite.

At the largest tested coefficient, $\lambda = 80.0$, the divergence rate is 0.0%. In this tabular Stag Hunt setup, high λ does not break by numerical explosion; a stronger counterexample must change the game geometry rather than only increasing the coefficient.

21 Phase V : payoff-geometry counterexample search

This phase searches a wider symmetric 2×2 payoff family with $R = 4$ and $P = 2$, varying the sucker payoff S and temptation payoff T while keeping (C, C) socially better than miscoordination. The goal is adversarial: find payoff geometries where the peer correction shrinks, rather than expands, the cooperative basin.

Meta-MAPG hurts PG in 35 searched payoff settings. These are not Stag-Hunt counterex-

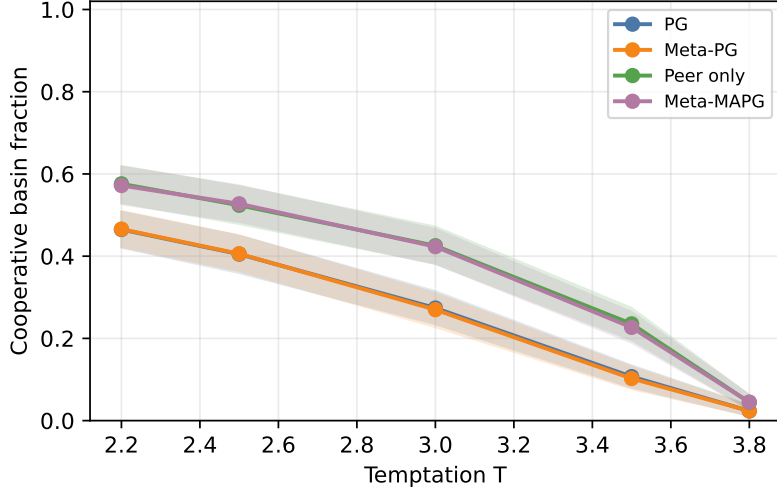


Figure 10: Cell-bootstrap T -sweep with all four methods. Meta-PG tracks PG, while Peer-only tracks Meta-MAPG, isolating the peer term as the source of basin expansion.

T	PG	Meta-PG	Peer only	Meta-MAPG
2.20	46.5 [42.1,50.8]%	46.6 [42.3,51.0]%	57.6 [53.0,61.9]%	57.2 [52.7,61.8]%
2.50	40.5 [35.9,45.0]%	40.6 [36.4,45.0]%	52.4 [47.8,57.0]%	52.7 [48.4,57.2]%
3.00	27.4 [23.5,31.5]%	27.0 [22.8,31.0]%	42.5 [38.2,47.2]%	42.3 [38.1,46.6]%
3.50	10.7 [8.1,13.4]%	10.2 [7.7,13.1]%	23.5 [19.6,27.4]%	22.7 [19.0,26.4]%
3.80	2.4 [1.2,3.7]%	2.4 [1.2,3.6]%	4.4 [2.8,6.2]%	4.5 [2.8,6.3]%

Table 8: Cell-bootstrap basin fraction intervals by temptation T .

amples to the main experiment; they are boundary cases showing why the theorem needs an explicit alignment condition instead of a universal peer-correction-improves-selection claim.

22 Phase W : separatrix-margin certificate

This phase turns the basin-entry story into a separatrix-margin test. Using the Phase B PG basin mask, each initial policy receives a signed distance to the nearest opposite PG-basin cell. For PG-fail cells this is the empirical distance that an early update must overcome before ordinary PG should be expected to finish the cooperative selection.

The median boundary distance is 0.051 for gained PG-fail cells and 0.308 for non-gained PG-fail cells. This is stricter than Phase R: the peer term must be large in the right direction relative to the local separatrix margin, not merely positive.

23 Phase X : cumulative early-alignment test

This phase checks whether the one-step alignment certificate remains valid over the finite early-time window that actually moves policies. For $K \in \{1, 5, 10, 25, 50\}$ we accumulate learning-rate-weighted Meta-MAPG prefix projections and ask whether the prefix predicts final Meta-MAPG gains among cells where PG fails.

The strongest peer-prefix AUC occurs at $K = 50$ with AUC 0.888. The largest gained-cell PG-entry rate occurs at $K = 50$ and is 88.6%. This tests a finite-time version of the theory: cumulative aligned motion should carry policies across the empirical PG separatrix.

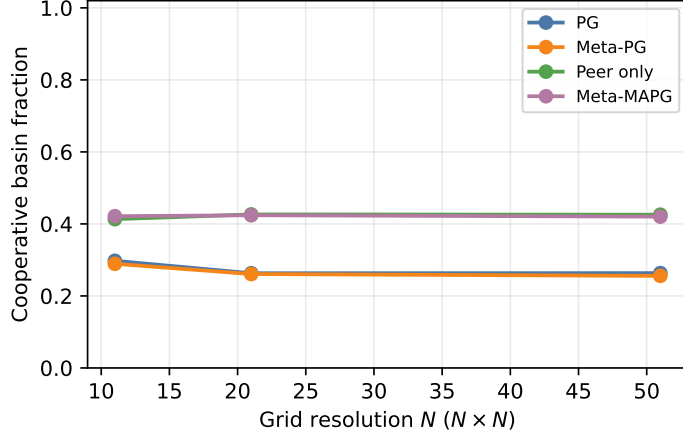


Figure 11: Basin fraction vs. grid resolution. All four methods stabilise by $N = 21$; ranking is preserved at all resolutions.

N	Peer only	Meta-MAPG	Meta-PG	PG
11	41.3%	42.1%	28.9%	29.8%
21	42.6%	42.4%	26.1%	26.3%
51	42.6%	42.0%	25.6%	26.3%

Table 9: Basin fraction vs. grid resolution.

24 Phase Y : minimal warm-up crossing time

This phase asks for the minimal Meta-MAPG warm-up needed to enter the ordinary PG basin. Starting from every Phase B grid cell where PG fails, we run exact Meta-MAPG updates and record the first step whose nearest grid cell lies inside the PG success basin.

This is the most direct operational form of the basin-entry claim: Meta-MAPG should only be needed until the trajectory crosses the ordinary PG separatrix. Cells that never cross within the tested window are candidates for either longer warm-up, weaker alignment, or a missing condition in the current theory.

25 Phase Z : adversarial alignment search

This phase tries to break the naive theory that a positive local peer projection is sufficient for basin improvement. For each payoff geometry from Phase V, we compare the average exact first-step peer displacement toward (C, C) against the final Meta-MAPG-minus-PG basin delta.

The worst searched payoff has $S = 0.50$ and $T = -1.00$, with basin delta -4.0%. The sweep finds 22 games where the all-cell average peer projection is positive but Meta-MAPG still shrinks the PG basin. It finds no such cases when the projection is averaged only over PG-fail cells. This breaks the broadest local-alignment claim while supporting the sharper margin/crossing condition from Phases W–Y.

26 Phase D2 : independent-seed ablation audit

Phase D used a shared pool of initial policies across all four arms. Here each arm draws its own independent initial policies to verify the conclusion is not a shared-seed artefact. This is a

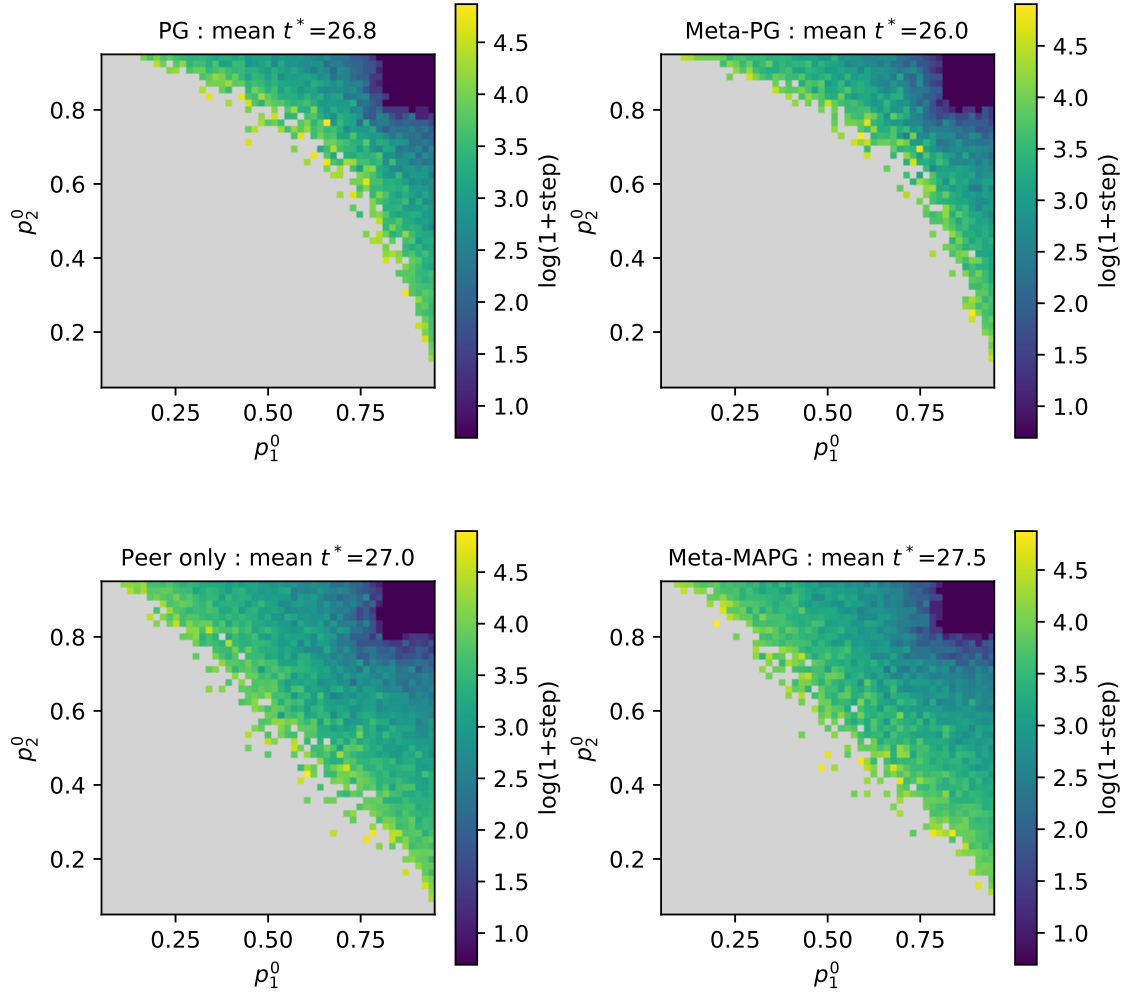


Figure 12: First-hit-time atlas ($\log(1 + t^*)$). Grey cells never reach the cooperative threshold. Peer-aware methods reach the threshold from a larger set of initial conditions; conditional hit times are not materially faster.

weaker, unpaired sanity check; the paired Phase D estimate remains the primary ablation. All three peer-aware arms again achieve higher point estimates than PG.

Method	Mean t^*	Frac. success
PG	26.8	26.3%
Meta-PG	26.0	25.9%
Peer only	27.0	42.4%
Meta-MAPG	27.5	42.3%

Table 10: First-hit-time summary. Mean computed over successful cells only.

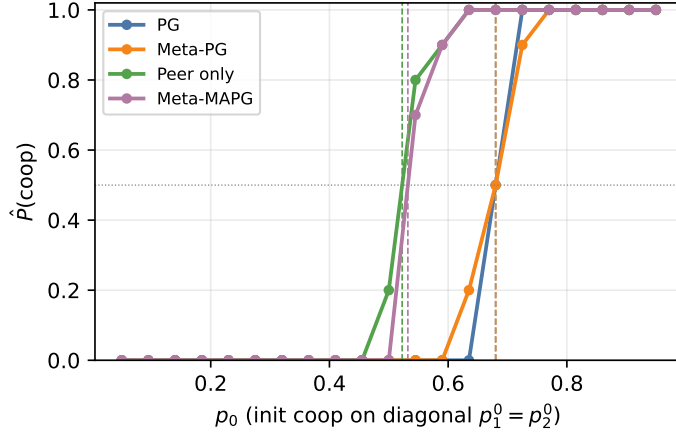


Figure 13: Success probability along the diagonal $p_1^0 = p_2^0$. Dashed verticals mark p_{diag}^* per method (crossing of the horizontal 0.5 line).

Method	p_{diag}^*
PG	0.680
Meta-PG	0.680
Peer only	0.522
Meta-MAPG	0.532

Table 11: Diagonal critical threshold: minimum initial cooperation on $p_1^0 = p_2^0$ for 50% basin-entry probability.

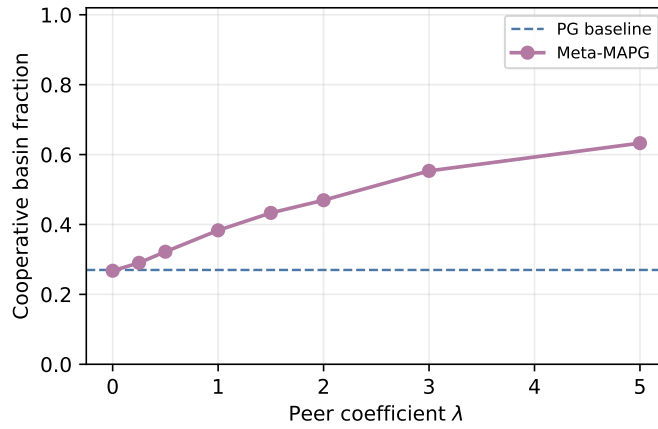


Figure 14: Peer-coefficient sweep. The cooperative basin expands once λ is positive and keeps increasing over the tested clipped tabular range; this sweep identifies coefficient sensitivity, not an asymptotically safe optimum. The $\lambda = 0$ point behaves like the own-only Meta-PG correction.

Method	λ	Coop. basin fraction
PG	–	27.0%
Meta-MAPG	0.00	26.8%
Meta-MAPG	0.25	29.0%
Meta-MAPG	0.50	32.2%
Meta-MAPG	1.00	38.3%
Meta-MAPG	1.50	43.3%
Meta-MAPG	2.00	46.9%
Meta-MAPG	3.00	55.3%
Meta-MAPG	5.00	63.3%

Table 12: Basin fraction as a function of peer coefficient.

Method	Basin	Expansion	Gained area	Lost area	Mean welfare
PG	25.9%	1.00×	0.0%	0.0%	4.99
Meta-PG	26.4%	1.02×	2.0%	1.6%	4.95
Peer only	42.9%	1.66×	17.0%	0.0%	5.62
Meta-MAPG	42.3%	1.63×	16.5%	0.0%	5.60

Table 13: Scalar summary metrics. Gained/lost area is measured relative to the PG basin mask. Welfare is $J_1 + J_2$ at the terminal policy, averaged over the welfare recomputation grid.

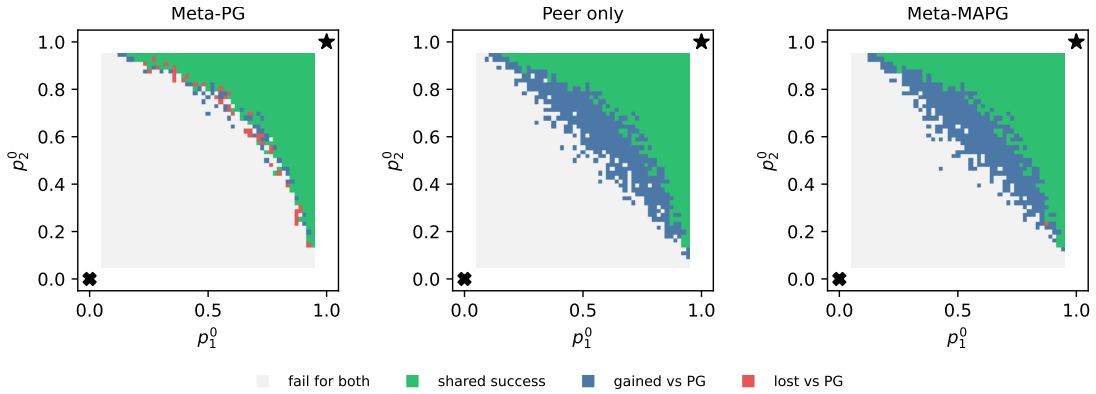


Figure 15: Basin gain/loss masks relative to PG. Peer-only and Meta-MAPG add a large contiguous region below the PG frontier, while Meta-PG adds almost nothing.

Method	Gained area	Lost area	Net area
Meta-PG	2.0%	1.6%	0.5%
Peer only	17.0%	0.0%	17.0%
Meta-MAPG	16.5%	0.0%	16.4%

Table 14: Area changes relative to PG on the Phase B grid.

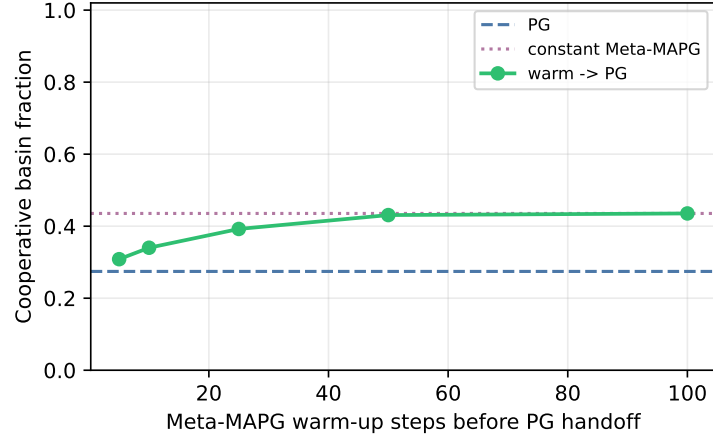


Figure 16: Warm-up handoff curve. Meta-MAPG is used only for the first K steps, followed by pure PG.

Schedule	Coop. basin fraction
PG only	27.4%
Constant Meta-MAPG	43.5%
5 warm steps $\rightarrow$ PG	30.8%
10 warm steps $\rightarrow$ PG	34.0%
25 warm steps $\rightarrow$ PG	39.2%
50 warm steps $\rightarrow$ PG	43.1%
100 warm steps $\rightarrow$ PG	43.5%

Table 15: Exact deterministic handoff basin fraction.

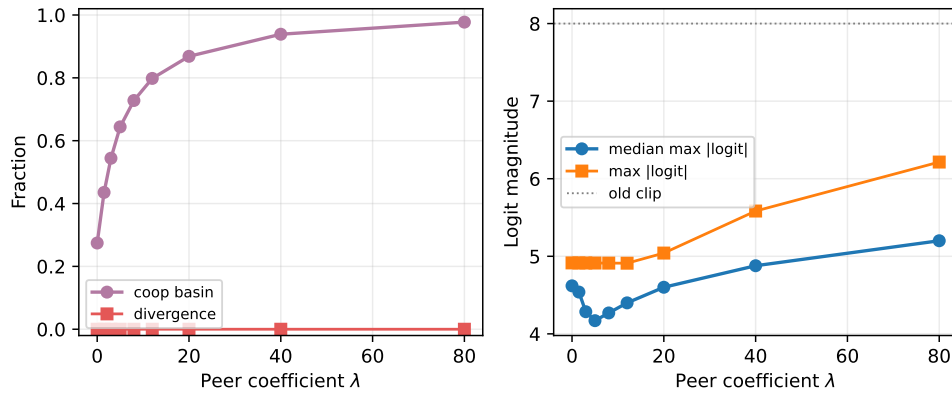


Figure 17: High- λ no-clip stress test in tabular Stag Hunt. The left panel tracks basin fraction and divergence; the right panel tracks logit magnitudes.

λ	Coop. basin	Divergence	Median max $ \theta $
0.0	27.4%	0.0%	4.62
1.5	43.5%	0.0%	4.54
3.0	54.4%	0.0%	4.29
5.0	64.4%	0.0%	4.17
8.0	72.8%	0.0%	4.27
12.0	79.8%	0.0%	4.40
20.0	86.8%	0.0%	4.60
40.0	93.9%	0.0%	4.88
80.0	97.7%	0.0%	5.20

Table 16: No-clip high-coefficient stress summary.

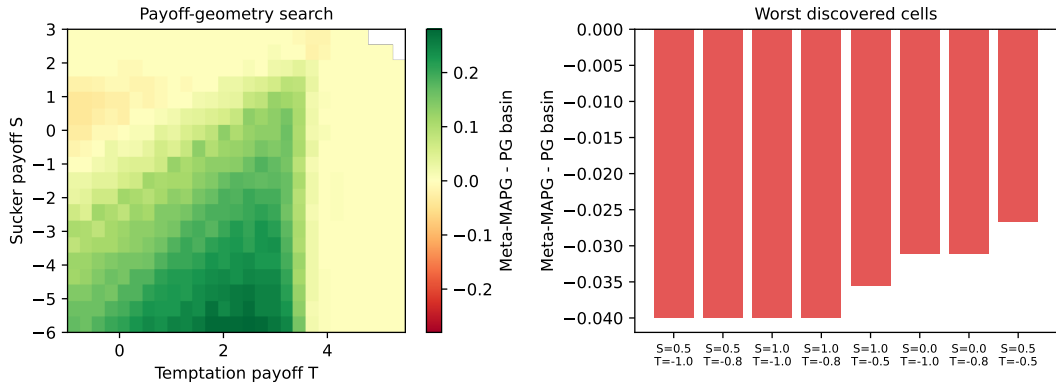


Figure 18: Payoff-geometry counterexample search. Green means Meta-MAPG expands the cooperative basin relative to PG; red means it shrinks it. The worst cases are harmony-like regions where PG already has a large cooperative basin.

Case	S	T	PG	Meta-MAPG	Δ
Worst	0.50	-1.00	84.4%	80.4%	-4.0%
Best	-6.00	2.50	8.0%	36.0%	28.0%

Table 17: Extremes from the payoff-geometry search.

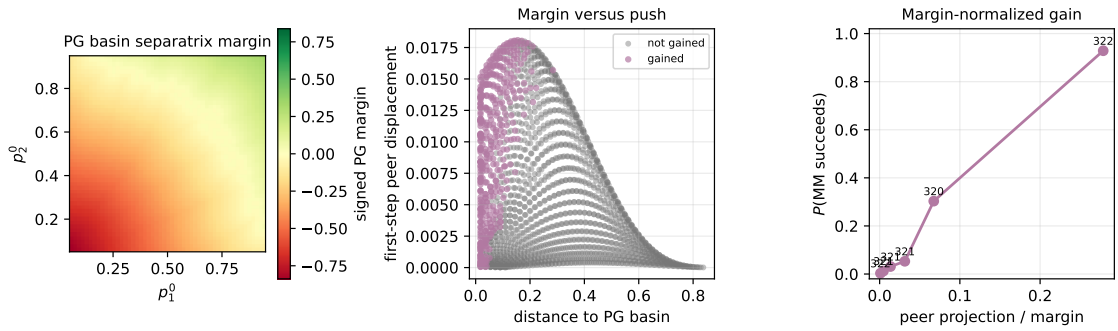


Figure 19: Separatrix-margin certificate. Left: signed distance to the PG basin boundary. Middle: first-step peer displacement versus boundary distance. Right: gain probability after normalizing the peer projection by the empirical boundary distance.

Metric	Peer / margin	Improvement / margin
AUC on PG-fail cells	0.954	0.959
Spearman with gain label	0.654	0.660

Table 18: Margin-normalized prediction of Meta-MAPG gains.

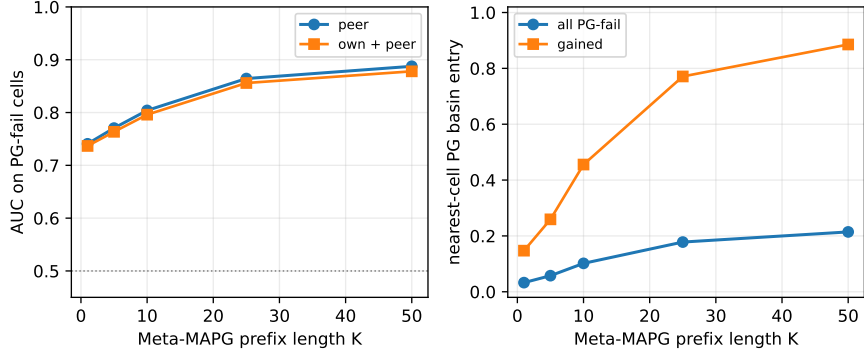


Figure 20: Cumulative early-alignment test. Left: AUC for predicting Meta-MAPG gains among PG-fail cells. Right: fraction of PG-fail cells whose Meta-MAPG prefix lands in the nearest-cell PG basin.

K	Peer AUC	Entry rate, PG-fail	Entry rate, gained
1	0.741	3.3%	14.7%
5	0.771	5.8%	25.9%
10	0.804	10.2%	45.6%
25	0.864	17.8%	77.1%
50	0.888	21.4%	88.6%

Table 19: Finite-prefix alignment and empirical PG-basin entry.

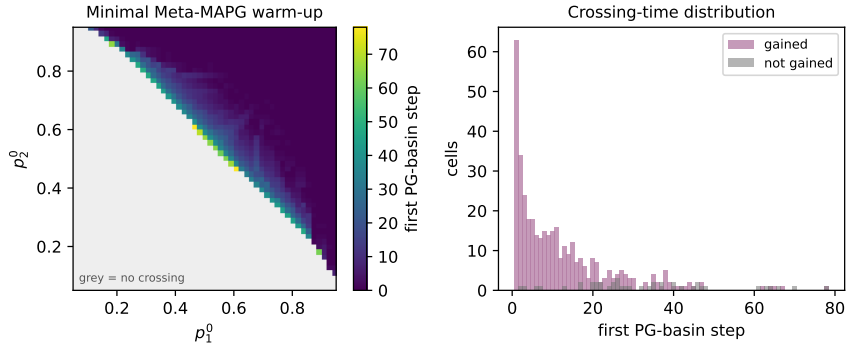


Figure 21: Minimal warm-up crossing time. Left: first Meta-MAPG step that reaches the empirical PG basin. Right: crossing-time distribution split by whether the cell is ultimately gained by Meta-MAPG.

Metric	Value
PG-fail cells	1927
Meta-MAPG gained cells	428
PG-fail cells crossing PG basin	427
Gained cells crossing PG basin	385
Crossing rate among gained cells	90.0%
Median crossing step among gained cells	8.0

Table 20: Minimal warm-up crossing summary.

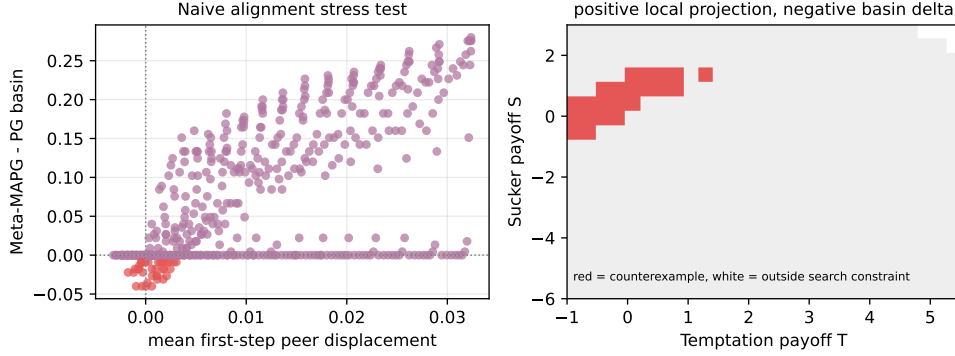


Figure 22: Adversarial alignment search. Left: mean first-step peer displacement versus final basin delta. Right: payoff cells where the mean local displacement is positive but Meta-MAPG still shrinks the PG basin.

Metric	Value
Games searched	509
Games where Meta-MAPG hurts PG	35
Positive all-cell projection but hurts	22
Positive PG-fail projection but hurts	0
Spearman(mean projection, basin delta)	0.654

Table 21: Adversarial test of a naive local-alignment theorem.

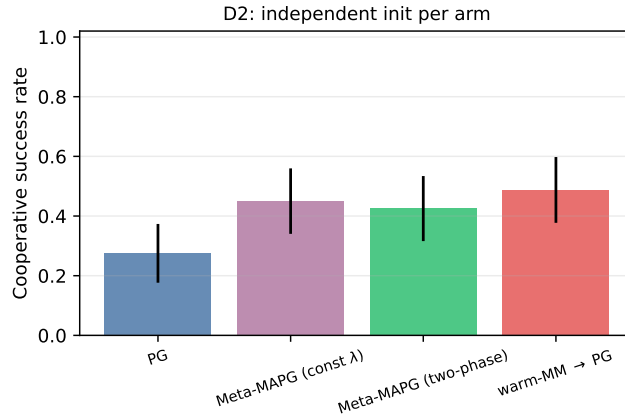


Figure 23: D2: same ablation as Phase D with independent initial policies per arm. This checks that Phase D is not an artefact of one shared seed pool, but it has lower power than the paired design.

Arm	Success rate	95% Wilson CI
PG	27.5%	[18.9%, 38.1%]
Meta-MAPG (const. λ)	45.0%	[34.6%, 55.9%]
Meta-MAPG (two-phase)	42.5%	[32.3%, 53.4%]
warm-Meta-MAPG $\rightarrow$ pure-PG	48.8%	[38.1%, 59.5%]

Table 22: D2 audit: independent initial conditions confirm the peer-aware point-estimate advantage is not caused by one shared seed pool.